Lab - Getting Familiar with the Linux Shell

Introduction

In this lab, you will use the Linux command line to manage files and folders, and perform some basic administrative tasks.

Part 1: Shell Basics

Part 2: Copying, Deleting, and Moving Files

Recommended Equipment

Security Workstation virtual machine

Instructions

Part 1: Shell Basics

The shell is the term used to refer to the command interpreter in Linux. Also known as Terminal, Command Line and Command Prompt, the shell is very powerful way to interact with a Linux computer.

Step 1: Access the Command Line

a. Log on to the Security Workstation VM as the **analyst** using the password **cyberops**. The account **analyst** is used as the example user account throughout this lab.

b. To access the command line, click the **terminal** icon located in the Dock, at the bottom of VM screen. The terminal emulator opens.



Step 2: Display Manual Pages from the command line.

You can display command line help using the **man** command. A man page, short for manual page, is a built-in documentation of the Linux commands. A man page provides detailed information about a given command and all its available options.

a. To learn more about the man page, type:

```
[analyst@secOps ~]$ man man
```

Name a few sections that are included in a man page.

b. Type **q** to exit the man page.

c. Use the **man** command to learn more about the **cp** command:

```
[analyst@secOps ~]$ man cp
```

What is the function of the **cp** command?

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What command would you use to find out more information about the **pwd** command? What is the function of the **pwd** command?

Step 3: Create and change directories.

In this step, you will use the change directory (**cd**), make directory (**mkdir**), and list directory (**ls**) commands.

Note: A directory is another word for folder. The terms directory and folder are used interchangeably throughout this lab.

a. Type **pwd** at the prompt.

```
[analyst@secOps ~]$ pwd
/home/analyst
```

What is the current directory?

b. Navigate to the **/home/analyst** directory if it is not your current directory. Type **cd /home/analyst**

```
[analyst@secOps ~]$ cd /home/analyst
```

c. Type **ls -l** at the command prompt to list the files and folders that are in the current folder. Standing for list, the **-l** option displays file size, permissions, ownership, date of creation and more.

```
[analyst@secOps ~]$ ls -l
total 20
drwxr-xr-x 2 analyst analyst 4096 Mar 22  2018 Desktop
drwxr-xr-x 3 analyst analyst 4096 Apr  2 14:44 Downloads
drwxr-xr-x 9 analyst analyst 4096 Jul 19  2018 lab.support.files
drwxr-xr-x 2 analyst analyst 4096 Mar 21  2018 second_drive
-rw-r--r-- 1 analyst analyst  255 Apr 17 16:42 space.txt
```

d. In the current directory, use the **mkdir** command to create three new folders: **cyops_folder1**, **cyops_folder2**, and **cyops_folder3**. Type **mkdir cyops_folder1** and press **Enter**. Repeat these steps to create **cyops_folder2** and **cyops_folder3**.

```
[analyst@secOps ~]$ mkdir cyops_folder1
[analyst@secOps ~]$ mkdir cyops_folder2
[analyst@secOps ~]$ mkdir cyops_folder3
[analyst@secOps ~]$
```

e. Type **ls -l** to verify that the folders have been created:

```
[analyst@secOps ~]$ ls -l
total 32
drwxr-xr-x 2 analyst analyst 4096 Aug 16 15:01 cyops_folder1
drwxr-xr-x 2 analyst analyst 4096 Aug 16 15:02 cyops_folder2
drwxr-xr-x 2 analyst analyst 4096 Aug 16 15:02 cyops_folder3
drwxr-xr-x 2 analyst analyst 4096 Sep 26  2014 Desktop
drwx------ 3 analyst analyst 4096 Jul 14 11:28 Downloads
drwxr-xr-x 8 analyst analyst 4096 Jul 25 16:27 lab.support.files
drwxr-xr-x 2 analyst analyst 4096 Mar  3 15:56 second_drive
-rw-r--r-- 1 analyst analyst  254 Aug 16 13:38 space.txt
```

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f. Type **cd /home/analyst/cyops_folder3** at the command prompt and press **Enter**.

```
[analyst@secOps ~]$ cd /home/analyst/cyops_folder3
[analyst@secOps cyops_folder3]$
```

Which folder are you in now?

Note: In the **[analyst@secOps ~]$** prompt above: The tilde symbol **~** represents the current user's home directory. In this example, the current user's home directory is **/home/analyst**. After the **cd /home/analyst/cyops_folder3** command, the current user's home directory is now **/home/analyst/cyops_folder3**.

Note: **$** (dollar sign) indicates regular user privilege. If a '**#'** (hashtag or pound sign) is displayed at the prompt, it indicates elevated privilege (**root user**).

Note: While these symbols, conventions and main concepts remain the same, the prompt of a terminal window is highly customizable in Linux. Therefore, the prompt structure seen in the Security Workstation VM will likely differ from the prompt in other Linux installations.

Challenge: Type the command **cd ~** and describe what happens.

Why did this happen?

g. Use the **mkdir** command to create a new folder named **cyops_folder4** inside the **cyops_folder3** folder:

```
[analyst@secOps ~]$ mkdir /home/analyst/cyops_folder3/cyops_folder4
[analyst@secOps ~]$
```

h. Use the **ls -l** command to verify the folder creation.

```
analyst@secOps ~]$ ls –l /home/analyst/cyops_folder3
total 4
drwxr-xr-x 2 analyst analyst 4096 Aug 16 15:04 cyops_folder4
```

i. Up to this point, we have been using *full or absolute paths.* Absolute path is the term used when referring to paths that always start at the root (/) directory. It is also possible to work with *relative paths.* Relative paths reduce the amount of text to be typed. To understand relative paths, we must understand the **.** and **..** (dot and double dot) directories. From the **cyops_folder3** directory, issue a **ls –la**:

```
analyst@secOps ~]$ ls –la /home/analyst/cyops_folder3
total 12
drwxr-xr-x  3 analyst analyst 4096 Aug 16 15:04 .
drwxr-xr-x 20 analyst analyst 4096 Aug 16 15:02 ..
drwxr-xr-x  2 analyst analyst 4096 Aug 16 15:04 cyops_folder4
```

The **-a** option tells **ls** to show all files. Notice the **.** and **..** listings shown by **ls**. These listings are used by the operating system to track the current directory (.) and the parent directory (..) You can see the use of the **.** and **..** when using the **cd** command to change directories. Using the **cd** command to change the directory to the **.** directory incurs no visible directory change as the **.** points to the current directory itself.

j. Change the current directory to /home/analyst/cyops_folder3:

```
[analyst@secOps ~]$ cd /home/analyst/cyops_folder3
[analyst@secOps cyops_folder3]$
```

k. Type **cd .**

```
[analyst@secOps cyops_folder3]$ cd .
[analyst@secOps cyops_folder3]$
```

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What happens?

l. Changing the directory to the **..** directory, will change to the directory that is one level up. This directory is also known as *parent directory*. Type **cd ..**

```
[analyst@secOps cyops_folder3]$ cd ..
[analyst@secOps ~]$
```

What happens?

m. What would be the current directory if you issued the **cd ..** command at [analyst@secOps ~]$?

What would be the current directory if you issued the **cd ..** command at [analyst@secOps home]$?

What would be the current directory if you issued the **cd ..** command at [analyst@secOps /]$?

Step 4: Redirect Outputs.

Another powerful command line operator in Linux is known as *redirect*. Represented by the **>** symbol, this operator allows the output of a command to be redirected to some location other the current terminal window (the default).

a. Use the **cd** command to change to the **/home/analyst/** (**~)** directory:

```
[analyst@secOps /]$ cd /home/analyst/
[analyst@secOps ~]$
```

b. Use the **echo** command to echo a message. Because no output was defined, echo will output to the current terminal window:

```
analyst@secOps ~]$ echo This is a message echoed to the terminal by echo.
This is a message echoed to the terminal by echo.
```

c. Use the **>** operator to redirect the output of echo to a text file instead of to the screen:

```
analyst@secOps ~]$ echo This is a message echoed to the terminal by echo. >
some_text_file.txt
```

No output was shown.

Is that expected? Explain.

d. Notice, that even though the **some_text_file.txt** file did not exist, prior to the echo command, it was automatically created to receive the output generated by **echo**. Use the **ls -l** command to verify if the file was really created:

```
[analyst@secOps ~]$ ls –l some_text_file.txt
-rw-r--r-- 1 analyst analyst   50 Feb 24 16:11 some_text_file.txt
```

e. Use the **cat** command to display the contents of the **some_text_file.txt** text file:

```
[analyst@secOps ~]$ cat some_text_file.txt
This is a message echoed to the terminal by echo.
```

f. Use the **>** operator again to redirect a different echo output of echo to the **some_text_file.txt** text file:

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```
analyst@secOps ~]$ echo This is a DIFFERENT message, once again echoed to the terminal by echo. > some_text_file.txt
```

g. Once again, use the **cat** command to display the contents of the **some_text_file.txt** text file:

```
[analyst@secOps ~]$ cat some_text_file.txt
This is a DIFFERENT message, once again echoed to the terminal by echo.
```

What happened to the text file? Explain.

Step 5: Redirect and Append to a Text File.

a. Similar to the **>** operator, the **>>** operator also allows for redirecting data to files. The difference is that **>>** appends data to the end of the referred file, keeping the current contents intact. To append a message to the some_text_file.txt, issue the command below:

```
[analyst@secOps ~]$ echo This is another line of text. It will be APPENDED to the output file. >> some_text_file.txt
```

b. Use the **cat** command to display the contents of the **some_text_file.txt** text file yet again:

```
[analyst@secOps ~]$ cat some_text_file.txt
This is a DIFFERENT message, once again echoed to the terminal by echo.
This is another line of text. It will be APPENDED to the output file.
```

What happened to the text file? Explain.

Step 6: Work with hidden files in Linux.

a. In Linux, files with names that begin with a '.' (single dot) are not shown by default. While dot-files have nothing else special about them, they are called hidden files because of this feature. Examples of hidden files are **.file5**, **.file6**, **.file7**.

Note: Do not confuse dot-files with the current directory indicator "**.**" symbol. Hidden file names begin with a dot (period), followed by more characters while the dot directory is a hidden directory comprised of only a single dot.

b. Use **ls -l** to display the files stored in the analyst home directory.

```
[analyst@secOps ~]$ ls –l
```

How many files are displayed?

c. Use the **ls -la** command to display all files in the home directory of analyst, including the hidden files.

```
[analyst@secOps ~]$ ls –la
```

How many more files are displayed than before? Explain.

Is it possible to hide entire directories by adding a dot before its name as well? Are there any directories in the output of ls -la above?

Give three examples of hidden files shown in the output of **ls -la** above.

d. Type the **man ls** command at the prompt to learn more about the **ls** command.

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```
[analyst@secOps ~]$ man ls
```

e. Use the down arrow key (one line at a time) or the space bar (one page at a time) to scroll down the page and locate the **-a** option used above and read its description to familiarize yourself with the **ls -a** command.

Part 2: Copying, Deleting, and Moving Files

Step 1: Copying Files

a. The **cp** command is used to copy files around the local file system. When using **cp**, a new copy of the file is created and placed in the specified location, leaving the original file intact. The first parameter is the source file and the second is the destination. Issue the command below to copy **some_text_file.txt** from the home directory to the **cyops_folder2** folder:

```
[analyst@secOps ~]$ cp some_text_file.txt cyops_folder2/
```

Identify the parameters in the cp command above.

What are the source and destination files? (use full paths to represent the parameters)

b. Use the **ls** command to verify that **some_text_file.txt** is now in **cyops_folder2:**

```
[analyst@secOps ~]$ ls cyops_folder2/
some_text_file.txt
```

c. Use the **ls** command to verify that **some_text_file.txt** is also in the home directory:

```
[analyst@secOps ~]$ ls -l
total 36
drwxr-xr-x 2 analyst analyst 4096 Aug 16 15:01 cyops_folder1
drwxr-xr-x 2 analyst analyst 4096 Aug 16 15:11 cyops_folder2
drwxr-xr-x 3 analyst analyst 4096 Aug 16 15:04 cyops_folder3
drwxr-xr-x 2 analyst analyst 4096 Sep 26  2014 Desktop
drwx------ 3 analyst analyst 4096 Jul 14 11:28 Downloads
drwxr-xr-x 8 analyst analyst 4096 Jul 25 16:27 lab.support.files
drwxr-xr-x 2 analyst analyst 4096 Mar  3 15:56 second_drive
-rw-r--r-- 1 analyst analyst  142 Aug 16 15:09 some_text_file.txt
-rw-r--r-- 1 analyst analyst  254 Aug 16 13:38 space.txt
```

Step 2: Deleting Files and Directories

a. Use the **rm** command to remove files. Issue the command below to remove the file **some_text_file.txt** from the home directory. The **ls** command is then used to show that the file **some_text_file.txt** has been removed from the home directory:

```
[analyst@secOps ~]$ rm some_text_file.txt
[analyst@secOps ~]$ ls -l
total 32
drwxr-xr-x 2 analyst analyst 4096 Aug 16 15:01 cyops_folder1
drwxr-xr-x 2 analyst analyst 4096 Aug 16 15:11 cyops_folder2
drwxr-xr-x 3 analyst analyst 4096 Aug 16 15:04 cyops_folder3
drwxr-xr-x 2 analyst analyst 4096 Sep 26  2014 Desktop
drwx------ 3 analyst analyst 4096 Jul 14 11:28 Downloads
```

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```
drwxr-xr-x 8 analyst analyst 4096 Jul 25 16:27 lab.support.files
drwxr-xr-x 2 analyst analyst 4096 Mar  3 15:56 second_drive
-rw-r--r-- 1 analyst analyst  254 Aug 16 13:38 space.txt
```

b. In Linux, directories are seen as a type of file. As such, the **rm** command is also used to delete directories but the **-r** (recursive) option must be used. Notice that all files and other directories inside a given directory are also deleted when deleting a parent directory with the -r option. Issue the command below to delete the **cyops_folder1** folder and its contents:

```
[analyst@secOps ~]$ rm –r cyops_folder1
[analyst@secOps ~]$ ls -l
total 28
drwxr-xr-x 2 analyst analyst 4096 Aug 16 15:11 cyops_folder2
drwxr-xr-x 3 analyst analyst 4096 Aug 16 15:04 cyops_folder3
drwxr-xr-x 2 analyst analyst 4096 Sep 26  2014 Desktop
drwx------ 3 analyst analyst 4096 Jul 14 11:28 Downloads
drwxr-xr-x 8 analyst analyst 4096 Jul 25 16:27 lab.support.files
drwxr-xr-x 2 analyst analyst 4096 Mar  3 15:56 second_drive
-rw-r--r-- 1 analyst analyst  254 Aug 16 13:38 space.txt
```

Step 3: Moving Files and Directories

a. Moving files works similarly to copying files. The difference is that moving a file removes it from its original location. Use the **mv** commands to move files around the local filesystem. Like the **cp** commands, the mv command also requires source and destination parameters. Issue the command below to move the **some_text_file.txt** from **/home/analyst/cyops_folder2** back to the home directory:

```
[analyst@secOps ~]$ mv cyops_folder2/some_text_file.txt .
[analyst@secOps ~]$ ls –l cyops_folder2/
total 0
[analyst@secOps ~]$ ls –l /home/analyst/
total 32
drwxr-xr-x 2 analyst analyst 4096 Aug 16 15:13 cyops_folder2
drwxr-xr-x 3 analyst analyst 4096 Aug 16 15:04 cyops_folder3
drwxr-xr-x 2 analyst analyst 4096 Sep 26  2014 Desktop
drwx------ 3 analyst analyst 4096 Jul 14 11:28 Downloads
drwxr-xr-x 8 analyst analyst 4096 Jul 25 16:27 lab.support.files
drwxr-xr-x 2 analyst analyst 4096 Mar  3 15:56 second_drive
-rw-r--r-- 1 analyst analyst  142 Aug 16 15:11 some_text_file.txt
-rw-r--r-- 1 analyst analyst  254 Aug 16 13:38 space.txt
```

What command did you use to accomplish the task?

Reflection

What are the advantages of using the Linux command line?

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